

DEPARTMENT OF MATHEMATICS
MAT111 INTRODUCTORY MATHEMATICS I
WEEK 7: LECTURE NOTE

RATIONAL FUNCTIONS

Definition 1. A *rational function* can be written in the form

$$f(x) = \frac{N(x)}{D(x)}$$

where $N(x)$ and $D(x)$ are polynomials and $D(x)$ is not the zero polynomial.

In general, the *domain* of a rational function of x includes all real numbers except x -values that make the denominator zero. Much of the discussion of rational functions will focus on the graphical behavior near the x -values that make the denominator zero.

NOTATIONS: Let a and b be real numbers and $f(x)$ be a real valued function.

- $x \rightarrow a^-$: x approaches to a from the left
- $x \rightarrow a^+$: x approaches to a from the right
- $x \rightarrow +\infty$: x increases without bound
- $x \rightarrow -\infty$: x decreases without bound
- $f(x) \rightarrow -\infty$ as $x \rightarrow a^-$: When x approaches to a from the left $f(x)$ decreases without bound.
- $f(x) \rightarrow -\infty$ as $x \rightarrow a^+$: When x approaches to a from the right $f(x)$ decreases without bound.
- $f(x) \rightarrow +\infty$ as $x \rightarrow a^-$: When x approaches to a from the left $f(x)$ increases without bound.
- $f(x) \rightarrow +\infty$ as $x \rightarrow a^+$: When x approaches to a from the right $f(x)$ increases without bound.
- $f(x) \rightarrow b$ as $x \rightarrow \infty$: When x increases without bound $f(x)$ approaches to b .
- $f(x) \rightarrow b$ as $x \rightarrow -\infty$: When x decreases without bound $f(x)$ approaches to b .

Example 2. Find the domain of $f(x) = \frac{1}{x}$ and discuss the behavior of f near any excluded x -values.

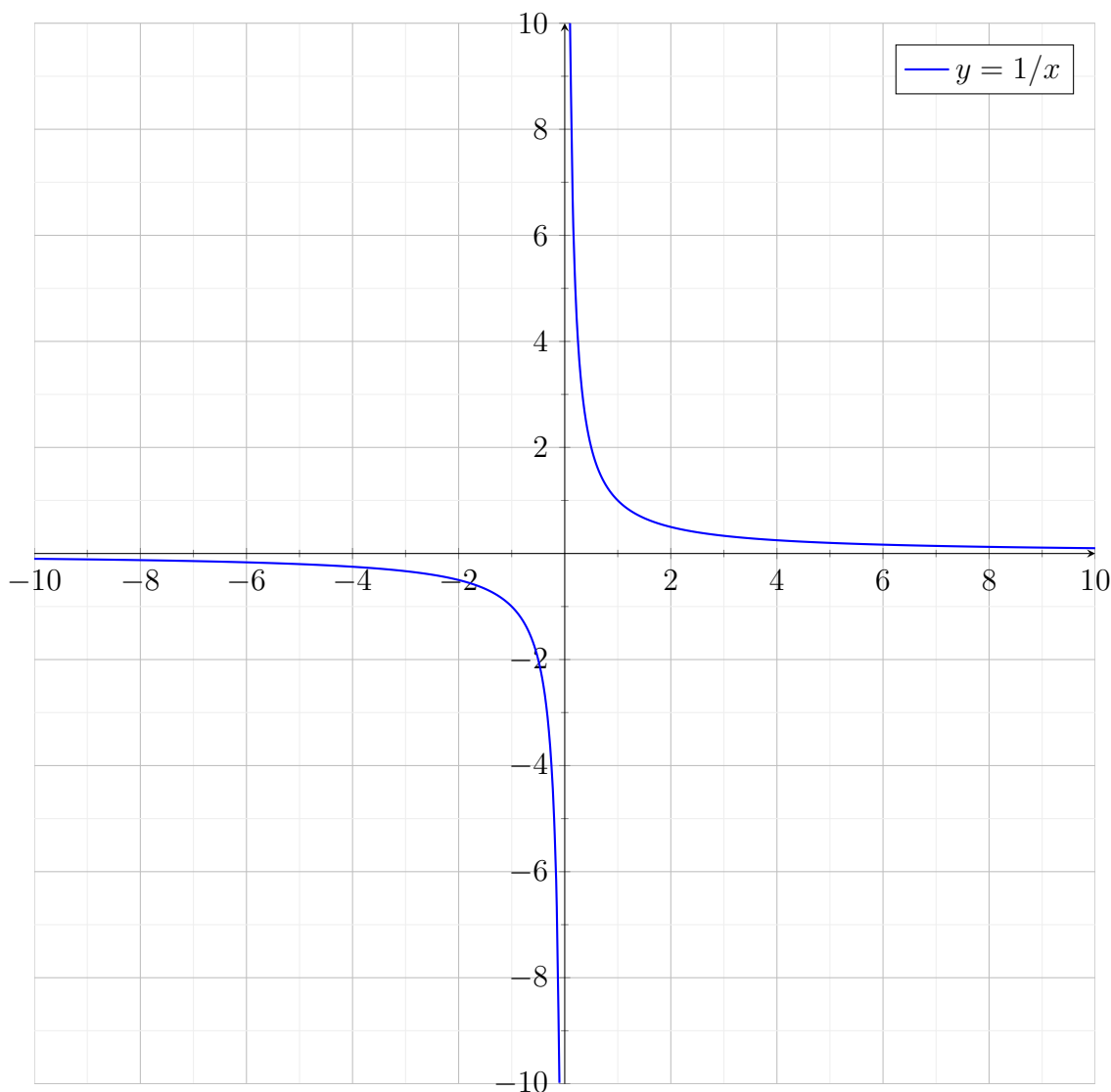
Solution. Because the denominator is zero when $x = 0$, the domain of f is all real numbers except $x = 0$, written $\mathbb{R} - \{0\}$ or $\mathbb{R} \setminus \{0\}$. To determine the behavior of f near this excluded value, evaluate $f(x)$ to the left and right of $x = 0$, as indicated below.

x	-1	-0.5	-0.1	-0.01	-0.001	-0.0001	$\rightarrow 0^-$
$f(x)$	-1	-2	-10	-100	-1000	-10,000	$\rightarrow -\infty$

From the table, we note that x approaches 0 from the left, $f(x)$ decreases without bound.

x	$0^+ \leftarrow$	0.000001	0.0001	0.001	0.01	0.1	0.5	1
$f(x)$	$\infty \leftarrow$	1,000,000	10,000	1000	100	10	2	1

From the table, we note that x approaches 0 from the right, $f(x)$ increases without bound. The graph of f is shown below.



Definition 3. (Asymptotes of the graph of a rational function)

1. The line $x = a$ is a **vertical asymptote** of the graph of f when

$$f(x) \rightarrow \infty \quad \text{or} \quad f(x) \rightarrow -\infty$$

as $x \rightarrow a$, either from the right or from the left.

2. The line $y = b$ is a **horizontal asymptote** of the graph of f when $f(x) \rightarrow b$ as $x \rightarrow \infty$ or $x \rightarrow -\infty$.

Let f be the rational function

$$f(x) = \frac{N(x)}{D(x)} = \frac{a_n x^n + a_{n-1} x^{n-1} + \dots + a_1 x^1 + a_0}{b_m x^m + b_{m-1} x^{m-1} + \dots + b_1 x^1 + b_0}$$

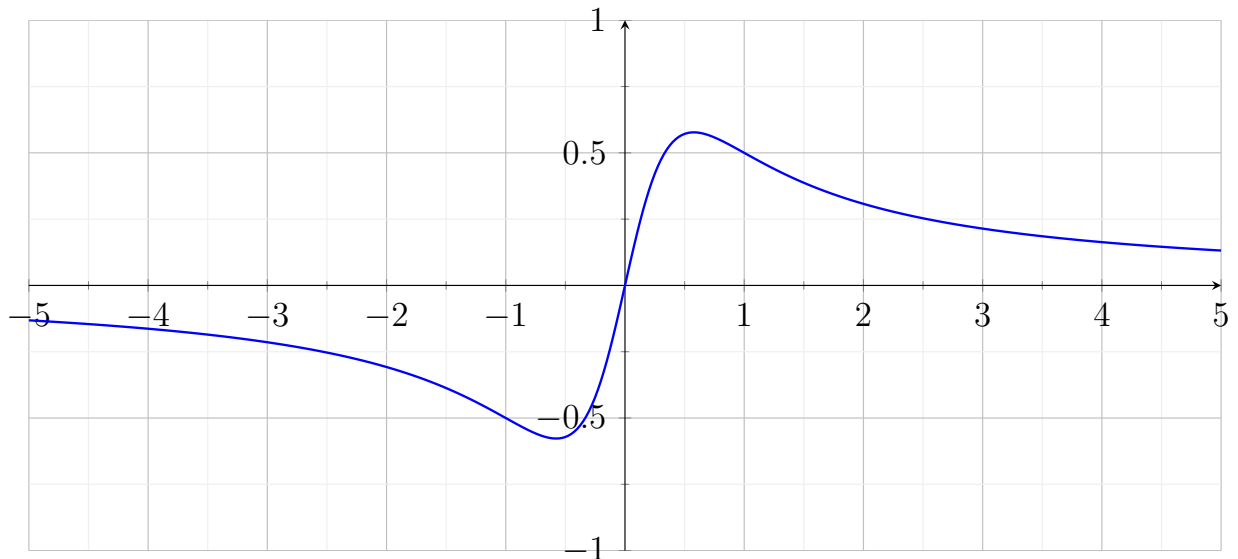
where $N(x)$ and $D(x)$ have **no common factors**.

- The graph of f has a vertical asymptotes at the solutions of $D(x) = 0$.
- The graph of f has at most one horizontal asymptotes determined by comparing the degrees of $N(x)$ and $D(x)$
 - ▷ If $n < m$, then the graph of f has the line $y = 0$ (the x -axis) as a horizontal asymptote.
 - ▷ If $n = m$, then the graph of f has the line $y = \frac{a_n}{b_m}$ as a horizontal asymptote, where a_n is the leading coefficient of the numerator and b_m is the leading coefficient of the denominator.
 - ▷ If $n > m$, then the graph of f has no horizontal asymptote.

Example 4. Find all asymptotes of the graph of each rational function.

1. $f(x) = \frac{2x}{3x^2 + 1}$.

Solution. Here the polynomials $N(x) = 2x$ and $D(x) = 3x^2 + 1$ have no common factors. For this rational function, the degree of the numerator is less than the degree of the denominator, so the graph has the line $y = 0$ as a horizontal asymptote. To find any vertical asymptote, set the denominator equal to zero and solve the resulting equation for x . Because the equation $3x^2 + 1 = 0$ has no real solutions, we can conclude that the graph has no vertical asymptotes. The graph of f is shown below.



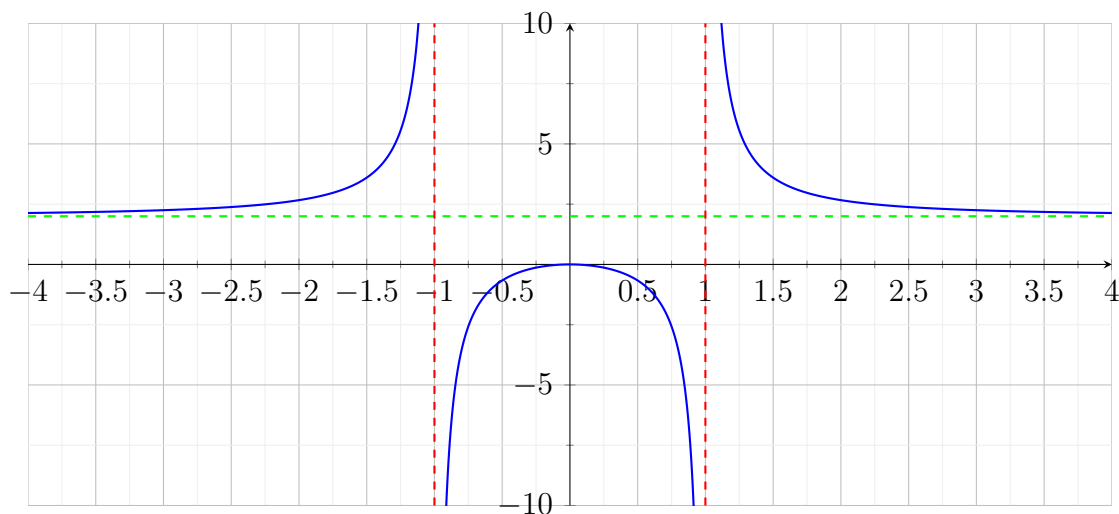
2. $f(x) = \frac{2x^2}{x^2 - 1}$.

Solution. Here the numerator $N(x) = 2x^2$ and the denominator $D(x) = x^2 - 1$ have no common factors. For this rational function, the degree of the numerator

is equal to the degree of the denominator. The leading coefficient of the numerator is 2 and the leading coefficient of the denominator is 1, so the graph has the line $y = \frac{2}{1} = 2$ as the horizontal asymptote. To find any vertical asymptotes, set the denominator equal to zero and solve the resulting equation for x .

$$\begin{aligned}x^2 - 1 &= 0 \\(x + 1)(x - 1) &= 0 \\x + 1 &\implies x = -1 \\x - 1 &= 0 \implies x = 1\end{aligned}$$

The graph has the lines $x = -1$ and $x = 1$ as vertical asymptotes.



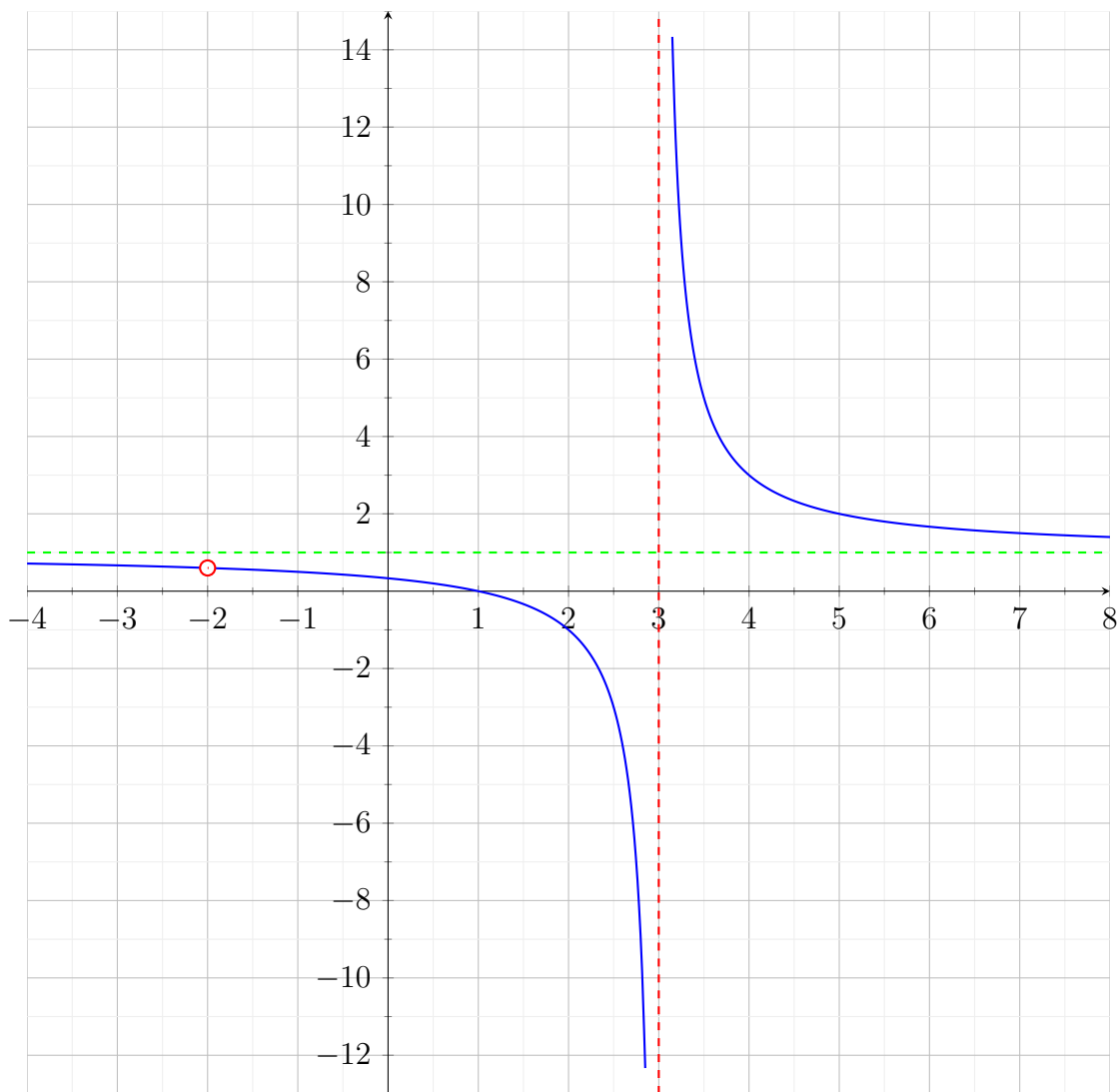
Example 5. Find all asymptotes and holes in the graph of

$$f(x) = \frac{x^2 + x - 2}{x^2 - x - 6}.$$

Solution: For this rational function the degree of the numerator is equal to the degree of the denominator. The leading coefficients of the numerator and denominator are both 1, so the graph has the line $y = 1/1 = 1$ as a horizontal asymptote. To find any vertical asymptotes first factor the numerator and denominator as follows.

$$\begin{aligned}f(x) &= \frac{x^2 + x - 2}{x^2 - x - 6} \\&= \frac{(x - 1)(x + 2)}{(x + 2)(x - 3)} \\&= \frac{x - 1}{x - 3}, \quad x \neq -2.\end{aligned}$$

The graph has the line $x = 3$ as a vertical asymptote. But, the function is undefined at $x = -2$ as well. There is a hole in the graph at $(-2, 0.6)$. The graph of f is shown below.



Example 6. Given the function

$$f(x) = \frac{3x^3 + 7x^2 + 2}{-2x^3 + 16},$$

find

- (a) the domain of f ,
- (b) The vertical asymptote of f , and
- (c) the horizontal asymptote.

Solution.

- (a) Because the denominator is zero when $-2x^3 + 16 = 0$, solve this equation to determine that the domain of f is all real numbers except $x = 2$ ($\mathbb{R} - \{2\}$ or $\mathbb{R} \setminus \{2\}$)
- (b) Because the denominator of f is zero at $x = 2$ and 2 is not a solution of the numerator, the graph of f has the vertical asymptote $x = 2$.
- (c) Because the degree of the numerator and denominator are the same, and the leading coefficient of the numerator is 3 and the leading coefficient of the denominator is -2, the horizontal asymptote of f is $y = -\frac{3}{2}$.

Guidelines to sketch the graph of a rational function

Let $f(x) = \frac{N(x)}{D(x)}$, where $N(x)$ and $D(x)$ are polynomials.

Step 1 Simplify f , if possible. Any restrictions on the domain of f not in the simplified function should be listed.

Step 2 Find and plot the y -intercept(if any) by evaluating $f(0)$.

Step 3 Find the solutions of the numerator (if any) by setting the numerator equal to zero. Then plot the corresponding x -intercepts.

Step 4 Find the solutions of the denominator (if any) by setting the denominator equal to zero. Then sketch the corresponding vertical asymptote using dashed vertical lines and plot the corresponding holes using open circles.

Step 5 Find and sketch any other asymptotes of the graph using dashed lines.

Step 6 Plot at least one point between and one point beyond each x -intercept and vertical asymptotes.

Step 7 Use smooth curves to complete the graph between and beyond vertical asymptotes, excluding any points where f is not defined.

Example 7. Sketch the graph of the function and describe how the graph is related to the graph of $f(x) = \frac{1}{x}$.

1. $g(x) = -\frac{1}{x+2}$

Solution. With respect to the graph of f , the graph of g is obtained by reflection in the x -axis followed by a horizontal shift two units to the left.

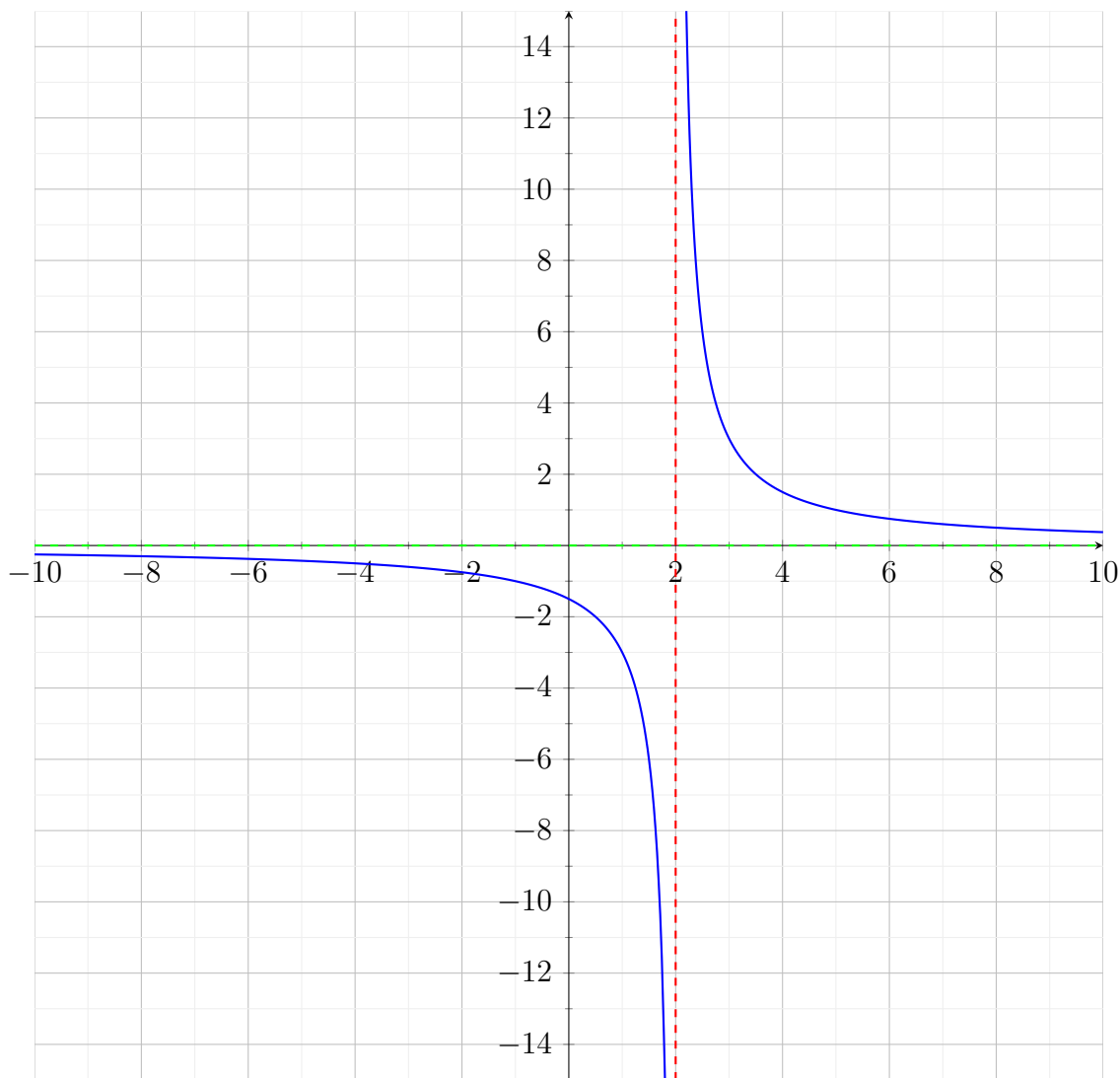
2. $h(x) = \frac{1}{x-1} + 3$

Solution. With respect to the graph of f , the graph of h is obtained by a horizontal shift one unit to the right followed by a vertical shift three units upward.

Example 8. Sketch the graph of $g(x) = \frac{3}{x-2}$ by hand.

- The point $\left(0, -\frac{3}{2}\right)$ is the y -intercept because $g(0) = -\frac{3}{2}$.
- There are no x -intercept because $3 \neq 0$.
- The line $x = 2$ is the vertical asymptote, the solution of $x - 2 = 0$.
- The line $y = 0$ is the horizontal asymptote because degree of $N(x) <$ degree of $D(x)$.
- Additional points

x	-4	1	1.5	1.8	1.95	1.99	2	2.01	2.05	2.2	2.5	3	5
$g(x)$	-0.5	-3	-6	-15	-60	-300	?	300	60	15	6	3	1



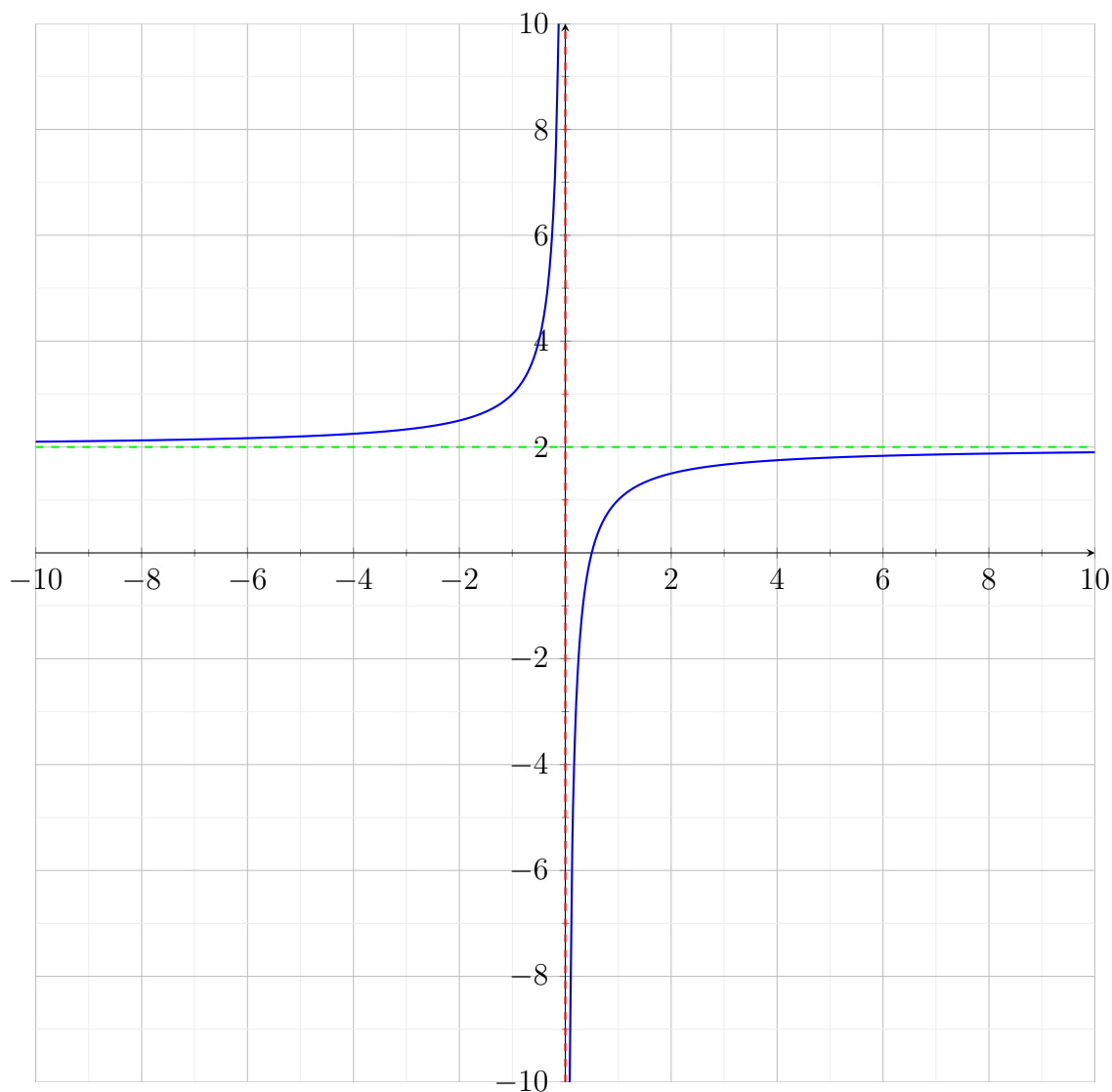
Note that the graph of g above is a vertical stretch and a right shift of the graph of $f(x) = \frac{1}{x}$ because

$$g(x) = \frac{3}{x-2} = 3 \left(\frac{1}{x-2} \right) = 3f(x-2).$$

Example 9. Sketch the graph of $f(x) = \frac{2x-1}{x}$ by hand.

- There are no y -intercepts, because $x = 0$ is not in the domain.
- The point $(\frac{1}{2}, 0)$ is the x -intercept, because $2x - 1 = 0$ when $x = \frac{1}{2}$.
- The line $x = 0$ is a vertical asymptote, solution of denominator.
- The line $y = 2$ is a horizontal asymptote, because degree of $N(x) =$ degree of $D(x)$.
- Additional points

x	-4	-1	0	1/4	4
$f(x)$	2.25	3	?	-2	1.75



Example 10. Sketch the graph of

$$f(x) = \frac{x}{x^2 - x - 2}$$

Solution. Factor the denominator to get

$$f(x) = \frac{x}{x^2 - x - 2} = \frac{x}{(x + 1)(x - 2)}.$$

Therefore, we note that

- The point $(0, 0)$ is the y -intercept, because $f(0) = 0$.
- The point $(0, 0)$ is the x -intercept, because the numerator is 0 when $x = 0$.
- The lines $x = -1$ and $x = 2$ are the vertical asymptotes because they are the zeros of the denominator.
- The line $y = 0$ is the horizontal asymptote, because degree of $N(x) < \text{degree of } D(x)$.

- *Additional points*

x	-3	-1	-0.5	1	2	3
$f(x)$	-0.3	?	0.4	-0.5	?	0.75



Example 11. *Sketch the graph of*

$$f(x) = \frac{x^2 - 9}{x^2 - 2x - 3}.$$

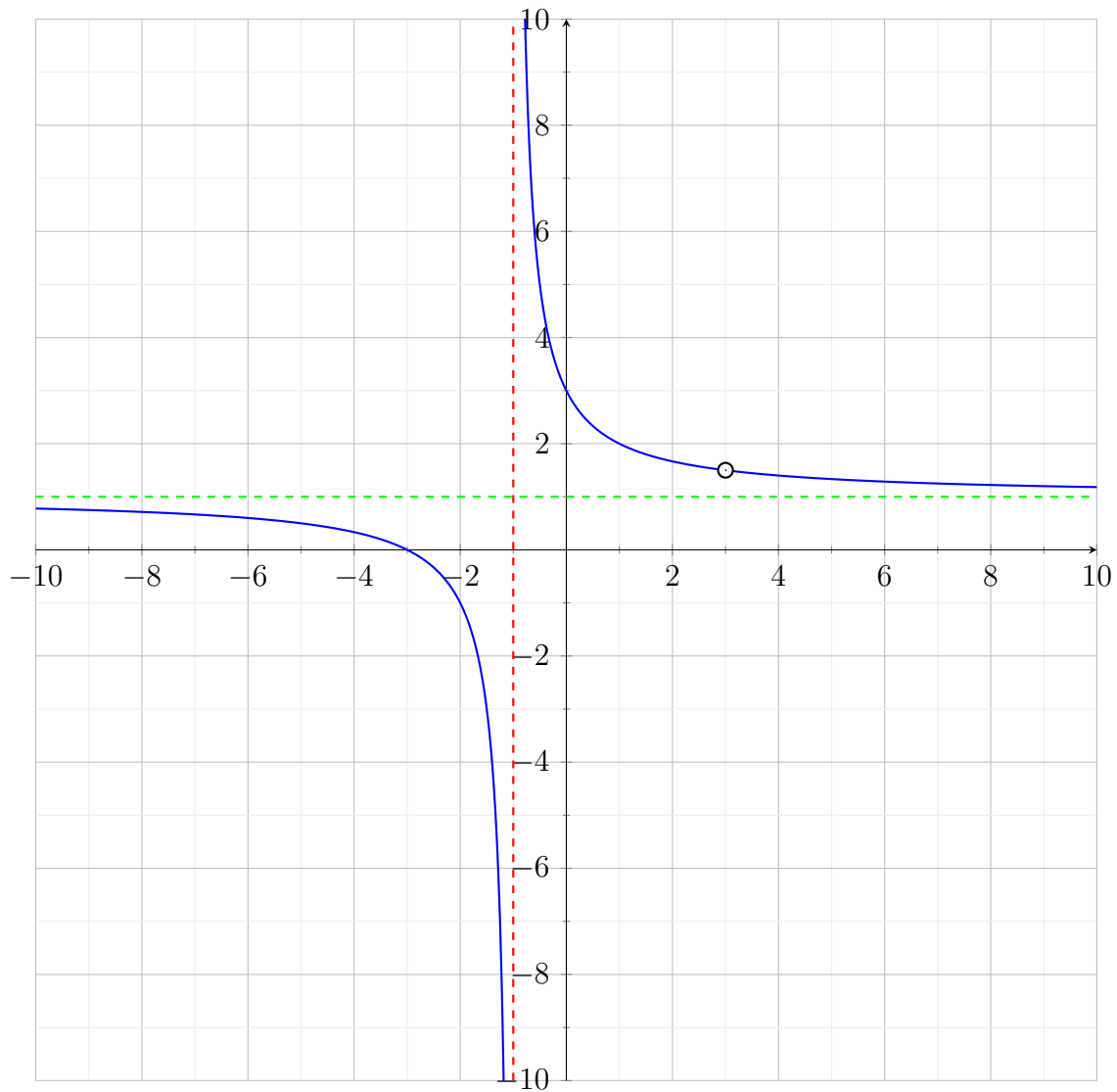
Solution. *Factorizing the numerator and denominator to get*

$$f(x) = \frac{x^2 - 9}{x^2 - 2x - 3} = \frac{(x - 3)(x + 3)}{(x - 3)(x + 1)} = \frac{x + 3}{x + 1}, \quad x \neq 3.$$

- *The point $(0, 3)$ is the y -intercept because $f(0) = 3$.*
- *The point $(-3, 0)$ is the x -intercept, because $x + 3 = 0$ when $x = -3$.*

- The line $x = -1$ is a vertical asymptote, the solution of denominator.
- The line $y = 1$ is a horizontal asymptote, because degree of $N(x) = \text{degree of } D(x)$.
- The point $(3, \frac{3}{2})$ is a hole, because f is not defined at $x = 3$.
- Additional points

x	-5	-2	-1	-0.5	1	3	4
$f(x)$	0.5	-1	?	5	2	?	1.4



Slant Asymptotes

Consider a rational function whose denominator is of degree 1 or greater. If the degree of the numerator is exactly one more than the degree of the denominator, then the graph of the function has a **slant (or oblique)** asymptote.

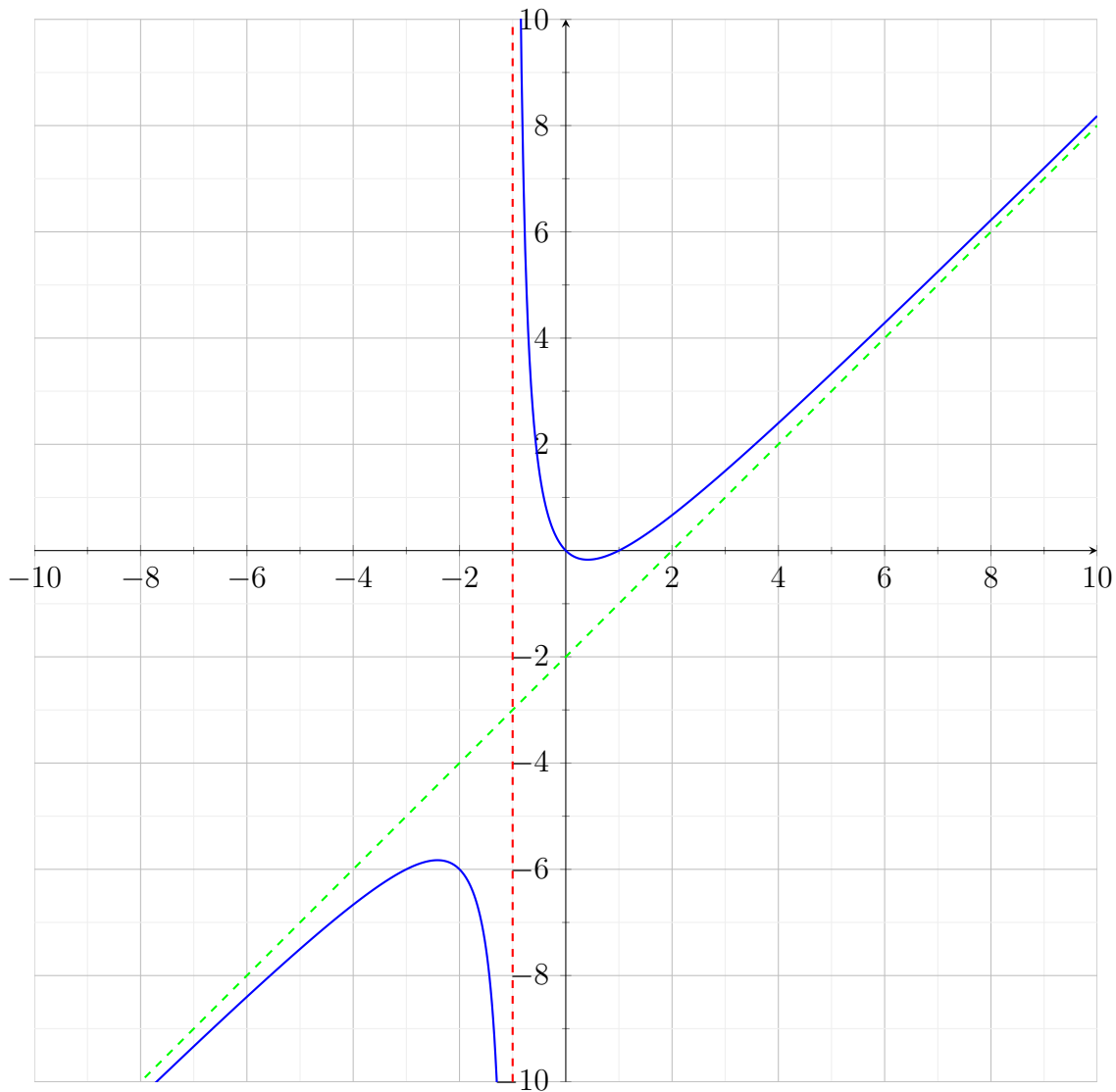
Example 12. The graph of

$$f(x) = \frac{x^2 - x}{x + 1}$$

has a slant asymptote as shown below.

Solution. To find the equation of a slant asymptote, use long division. For instance, by dividing $x + 1$ into $x^2 - x$ we have

$$f(x) = \frac{x^2 - x}{x + 1} = x - 2 + \frac{2}{x + 1}.$$



So, the graph of f approaches the line $y = x - 2$.

Example 13. Sketch the graph of

$$f(x) = \frac{x^2 - x - 2}{x - 1}.$$

Solution. Write in two ways

$$(i) \quad f(x) = \frac{x^2 - x - 2}{x - 1} = \frac{(x - 2)(x + 1)}{x - 1}$$

$$(ii) \quad f(x) = \frac{x^2 - x - 2}{x - 1} = x - \frac{2}{x - 1}.$$

- The point $(0, 2)$ is an y -intercept, because $f(0) = 2$.
- The points $(-1, 0)$ and $(2, 0)$ are the x -intercepts, because $f(-1) = 0$ and $f(2) = 0$.
- There are no horizontal asymptote, because degree of $N(x) >$ degree of $D(x)$.
- The line $y = x$ is the slant asymptote.
- Additional points

x	-2	0.5	1	1.5	3
$f(x)$	-1.33	4.5	?	-2.5	2

